Continued Fractions

$$a_0 + \cfrac{1}{a_1 + \cfrac{1}{a_2 + \cfrac{1}{a_3 + \cfrac{1}{a_4 + \cdots}}}}$$

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Notation

$$a_0 + \cfrac{1}{a_1 + \cfrac{1}{a_2 + \cfrac{1}{a_3 + \cfrac{1}{a_4 + \cdots}}}} = [a_0\,;a_1, a_2, a_3, a_4] = [a_0, a_1, a_2, a_3, a_4]$$

Any number can be written in form of a continued fraction.

Interesting fact

The continued fractions of golden ratio is: [1,1,1,1,...]

1. Finite
2. Infinite

1. Simple form
2. General form

Example: $\frac{348}{72} \approx 4.8333$

$\frac{348}{72} = 4 + \frac{60}{72}$	[4]	4	$\frac{348}{72} \approx 4$	First Convergent
$\frac{72}{60} = 1 + \frac{12}{60}$	[4,1]	$4 + \frac{1}{1}$	$\frac{348}{72} \approx 5$	Second Convergent
$\frac{60}{12} = 5$	[4,1,5]	$4 + \frac{1}{1 + \frac{1}{5}}$	$\frac{348}{72} \approx 4.8333$	Third Convergent

It is like Approximating a number!

Origins

Euclid's GCD algorithm

1. If $a < b$, swap a and b
2. Divide a by b and find the reminder r.
   if $r = 0$, then GCD is b
3. If $r \neq 0$, then set $a = b$, $b = r$ and go back to 1



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Example:

$a = 43,\ b = 19$

$$43 = \mathbf{2} \times 19 + 5$$
$$19 = \mathbf{3} \times 5 + 4$$
$$5 = \mathbf{1} \times 4 + 1$$
$$4 = \mathbf{4} \times 1 + 0$$

$$\frac{43}{19} = 2 + \cfrac{1}{3 + \cfrac{1}{1 + \cfrac{1}{4}}}$$

Some Properties

Rational Numbers

Rational numbers can be expressed as a finite Continued fraction and vice versa.

Inverting a fraction

If $P = [a_0, a_1, a_2, a_3, a_4]$ then $\frac{1}{P} = [0, a_0, a_1, a_2, a_3, a_4]$

If $P = [0, a_0, a_1, a_2, a_3, a_4$ then $\frac{1}{P} = [a_0, a_1, a_2, a_3, a_4]$

Multiple CF

$A = [a_0, a_1, a_2, a_3, \dots, a_n] = [a_0, a_1, a_2, a_3, \dots, a_n - 1, 1]$

Because $\frac{1}{a_n} = \frac{1}{a_n - 1 + \frac{1}{1}}$

If the last term is 1 we will add it to the previous term.

Uniqueness of CF

$x = [a_0, a_1, a_2, a_3, \dots, a_n]$

$x = [b_0, b_1, b_2, b_3, \dots, b_N]$

and a_n>1 and b_N>1 then n=N the fractions are same

Some Applications

Number Theory

Physics

Analysis

Cryptography

Algebra

Quadratic equations

$ax^2 + bx + c = 0$

Examples

The usual method yield, in general, irrational answers; therefore, we refer to CF method to get approximate roots in a rational form.

Simple form

$$x^2 - 2 = 0$$
$$x^2 - 1 = 1$$
$$(x-1)(x+1) = 1$$
$$x - 1 = \frac{1}{1+x}$$
$$x = 1 + \frac{1}{1+x}$$
$$x = 1 + \frac{1}{1 + (1 + \frac{1}{1+x}}$$
$$x = 1 + \frac{1}{2 + \frac{1}{x}}$$
$$x = [1, 2, 2, 2, \dots]$$

General form

$$x^2 - 5x - 6 = 0$$
$$x - 5 - \frac{6}{x} = 0$$
$$x = 5 + \frac{6}{x}$$
$$x = 5 + \frac{6}{5 + \frac{6}{x}}$$

And so on…

Hermite differential equation

$$\ddot{y} - 2x\dot{y} + 2\alpha y = 0$$

$$\frac{\mathrm{d}\ln y}{\mathrm{d}x} = \frac{\dot{y}}{y} = \frac{2\alpha}{2x - \frac{\ddot{y}}{\dot{y}}}$$

Differentiating the first equation and few simple steps

$$\frac{\mathrm{d}\ln y}{\mathrm{d}x} = \frac{\dot{y}}{y} = \frac{2\alpha}{2x - \frac{2(\alpha-1)}{2x - \frac{\dddot{y}}{\ddot{y}}}}$$

Again differentiating the equation twice we get

$$\frac{\mathrm{d}\ln y}{\mathrm{d}x} = \frac{\dot{y}}{y} = \frac{2\alpha}{2x - \frac{2(\alpha-1)}{2x - \frac{2(\alpha-2)}{2x - \frac{y^{iv}}{\dddot{y}}}}}$$

Now α can be α = 0, 1, 2, 3, …

$$\alpha = 0 \quad \rightarrow y = c$$
$$\alpha = 1 \quad \rightarrow y = c\mathrm{x}$$
$$\alpha = 2 \quad \rightarrow y = c(2x^2 - 1)$$

These polynomials are first few Hermite polynomials with a appropriate c

Finding roots of real numbers

Method 1

To get the square root of a certain number A and if a is the largest number such that its square is less than A; then we can compute the square root of A as shown in the example below

Example

$$Here\ A = \sqrt{2}\ and\ a = 1$$

$$\sqrt{2} = 1 + \sqrt{2} - 1$$

$$\sqrt{2} = 1 + \frac{1}{\frac{1}{\sqrt{2}-1}}$$

$$\sqrt{2} = 1 + \frac{1}{\frac{1}{\sqrt{2}-1} \times \frac{\sqrt{2}+1}{\sqrt{2}+1}}$$

$$\sqrt{2} = 1 + \frac{1}{1+\sqrt{2}}$$

$$\sqrt{2} = 1 + \frac{1}{1+1+\frac{1}{1+\sqrt{2}}}$$

$$\sqrt{2} = 1 + \frac{1}{2+\frac{1}{1+\sqrt{2}}}$$

$$\sqrt{2} = [1, 2, 2, 2, \ldots]$$

Because of limited time, I skipped the two other methods and the method of finding n^{th} root.

Comparison between the Use of CF and Some Other Numerical Methods in Computing Roots

Number	CFM	GM	NRM
$\sqrt{2}$	1.4142136248	1.41421356237	1.41421356
$\sqrt[3]{56423456}$	383.54815	383.54815	383.54815

General Method (GM) is slow but exact

NRM > CFM > GM

CF 10 steps for $\sqrt{2}$ and 5 steps for $\sqrt[3]{56423456}$
NRM 5 steps for both

Computation

```
function ContinuedFraction(x::Number,n::Int)
    global theNumber = BigFloat(x)
    global indices = []

    for i in 1:n
        push!(indices,Int64((round(theNumber,RoundDown))))
        if theNumber - round(theNumber,RoundDown) > 1e-12
            theNumber = 1 / (theNumber - round(theNumber,RoundDown))
        else theNumber = 0
            break
        end
    end
    return indices
end
```

Thank you for your attention

Reference

Ibran, Z.M., Aljatlawi, E.A. and Awin, A.M. (2022) On Continued Fractions and Their Applications. Journal of Applied Mathematics and Physics, 10, 142-159.